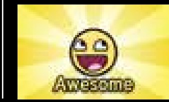
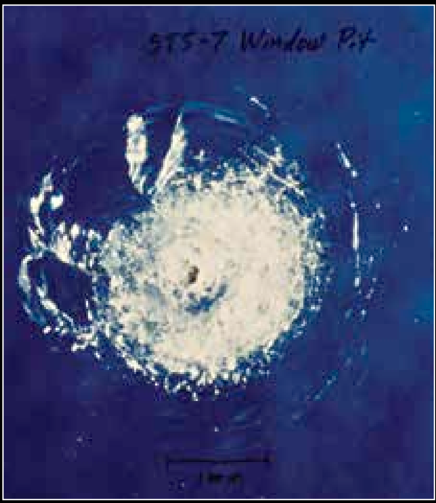
Sentient bots impossible
If you do the math, sentient robots aren't a mathematical possibility or probability. Here's why: **http://dgit.in/TFY6Ge**



The world's awesome
And we prove it to you by highlighting 60 things of the world's awesomeness: **http://dgit.in/1m5QVil**



The damage shown is on the Space Shuttle Challenger's front window and was caused by a fleck of paint.

munications satellite careening it out of its orbit and possibly crashing it into another satellite. This is not a movie reality but a real possibility.

The very reason the orbits are cluttered is because of the prime satellite real estate that exists between 200 and 2,000 kilometers for useful satellite placement. And it is there that our spacecrafts have ejected most of the waste to begin with - from rocket boosters to human waste and everything else - it's like littering the road outside your house. The extent of this waste land is across all orbit ranges from 200 kilometer to 36,000 kilometers.

All of these orbit bands are cluttered to varying degrees. The lower regions which are ideal for piloted spacecrafts and the 10,000 to 20,000 kilometer range orbits which are best for navigation and communication satellites - both are under siege of space junk. The highest band used for geosynchronous telecommunications and weather satellites is the sparsely polluted orbit purely due to its relatively rare usage.

However, the most critically damaging areas for collisions are within the Low Earth Orbits, which are sun-synchronous, as well as the higher altitude orbits which are geostationary. Both of these areas are critical to our presence in space and have been seriously cluttered over time.

And even though we've seen smaller and mid-sized fragments in the lower orbits quickly descend into the atmosphere and burn up - there are still more than 1,500 objects that are larger than 100 kilograms in mass moving at velocities of nearly 8 kilometers per second, directly crossing paths with all our essential satellite orbits. Together with the hundreds of thousands of mid-sized objects and the millions of small-sized ones, we can expect to see the Earth being slowly enveloped in a field of trash in the next few decades, which is a clear and present threat to our relationship with space travel.

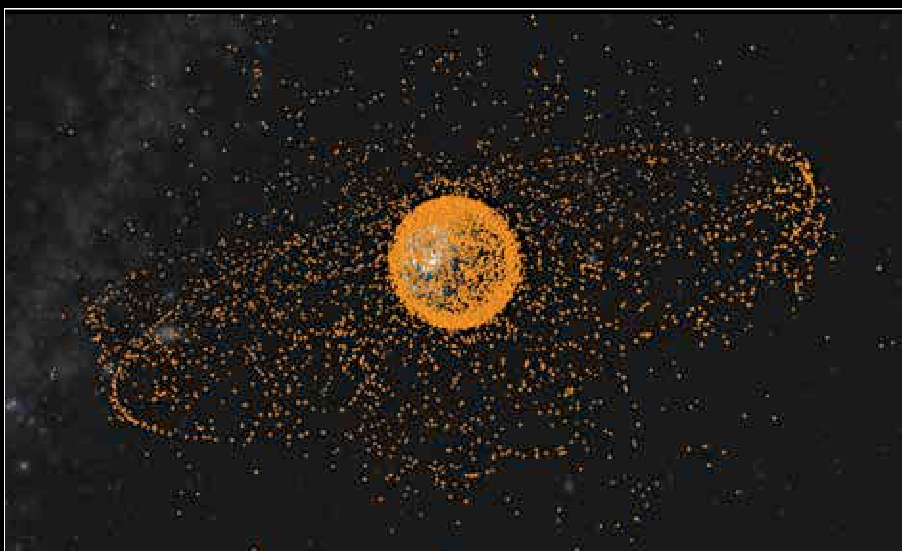
Orbital Collisions and the Kessler Syndrome

The two most recent events that gave light to this threat of orbital collisions took place in the last decade. The first incident was a large collision that took place in 2007, when China's botched attempt to destroy an old weather satellite with missiles ended up creating 150,000 new fragments. One of these fragments was the prime suspect of causing the 2013 collision with a Russian BLITS (Ball Lens In the Space) laser ranging satellite.

The second incident was the first real major collision of space debri and took place in 2009 when a 950 kilogram defunct Russian satellite - the Kosmos-2251 - collided with a 560 kilogram operational commercial American satellite - the Iridium-33 - just 800 kilometers over the Earth (Serbia) at relative speeds of over 42,000 kilometers per hour. This epic catastrophe scattered incalculable fragments of space trash in heaping clouds over the Earth endangering all known space programs in the area. These threats have gone so far that the International Space Station has had to radically adjust its orbit path twice in one month (April 2014) to avoid a collision with large pieces of space junk.

This increasing threat has been researched extensively by many scientists in recent years but the most significant warnings were given by NASA scientist Donald J. Kessler early in the history of space travel. In 1978, Kessler conceived a scenario where the density of debri in Low Earth Orbit would reach such critical mass that a cascade of collisions would be inevitable. Kessler updated his work by 1991 with new data and measurements making it definitively clear that a 1 kilogram fragment could easily destroy a 1000 kilogram spacecraft into numerous 1 kilogram pieces. This exponential growth in large sized space debris has been modeled and studied to be surprisingly accurate to became popularly known as the Kessler Syndrome

The sequence of collisions brought on by the Kessler Syndrome is predicted fill up the Low Earth Orbit region with millions of pieces of shrapnel which would make it impossible to navigate future space missions due to the resulting wall of garbage. And as more and more equipment finds its way to orbit, the likelihood that the Kessler Syndrome type outcome might occur becomes even greater.



We might have just invented a fence to separate us from space by sheer stupidity.